13) $\sqrt{3}\left(\cos \frac{\pi}{6} + i\sin \frac{\pi}{6}\right)$

14) $\sqrt{30}(\cos \pi + i\sin \pi)$

**Simplify. Write your answer in rectangular form when rectangular form is given and in polar form when polar form is given.**

15) $(4 + 4i)(5 - 3i)$

16) $4\sqrt{2}\left(\cos \frac{7\pi}{4} + i\sin \frac{7\pi}{4}\right) \cdot 2\left(\cos \frac{\pi}{6} + i\sin \frac{\pi}{6}\right)$

17) $\frac{6 - 2i}{2 + 4i}$

18) $\dfrac{2\sqrt{6}\left(\cos \frac{7\pi}{6} + i\sin \frac{7\pi}{6}\right)}{6\left(\cos \frac{11\pi}{6} + i\sin \frac{11\pi}{6}\right)}$

19) $(-1 - 6i)^3$

20) $\left(2\left(\cos \frac{7\pi}{6} + i\sin \frac{7\pi}{6}\right)\right)^3$

**Find all *n*th roots. Write your answers in rectangular form when rectangular form is given and in polar form when polar form is given.**

21) $2i,\ n = 3$

22) $6\left(\cos \frac{3\pi}{4} + i\sin \frac{3\pi}{4}\right),\ n = 4$

**Critical thinking questions:**

23) Show that $-i\sqrt[23]{n}$ is a 23rd root of $ni$.

24) Solve for $x : (2 + i)x = 1 + 2i$
Hint: $x$ is a complex number.